

Data deep-dive research

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Short recap : ViewSpatial-Bench

- Motivation : VLM understands only camera-perspective and lacks allocentric spatial reasoning
 - -> Train with allocentric data and fine tune it!
 - The fine tuned model called “MVSM” appears to understand the allocentric spatial reasoning

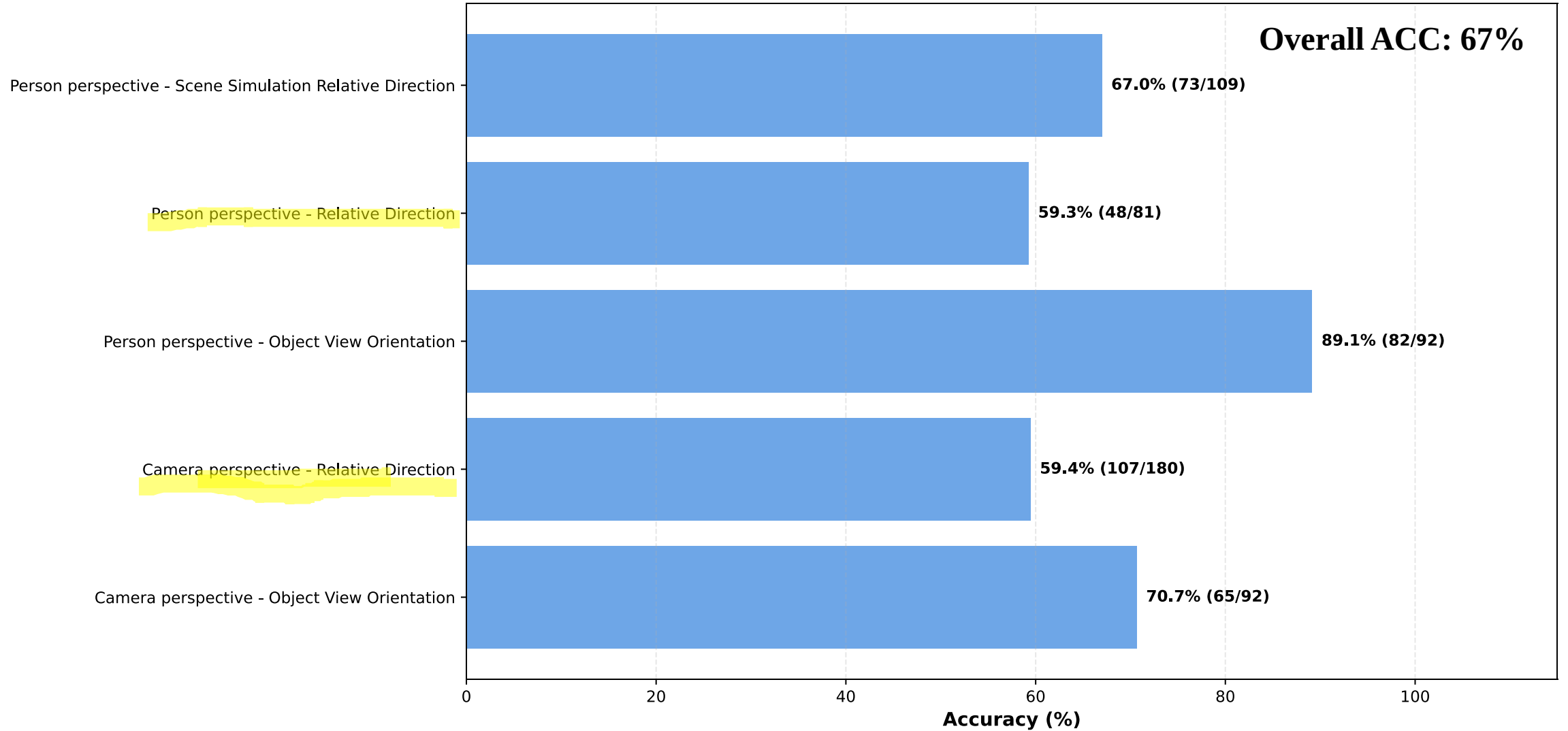
Short recap

- Problems with the last research
 - Haven't looked into the data in detail.
 - A one-to-one comparison of a different model(MVSM) from the paper

Introduction

- Is there a task that you get more wrong than others? Why?
- If you get VQA wrong, is there a tendency for the model to give incorrect answers?
 - Left -> Right? Or is it completely wrong?
- Baseline : Vanilla MVSM (fine tuned Qwen2.5)

Task-wise Accuracy (Baseline)



Hypothesis 1

Data Augmentation

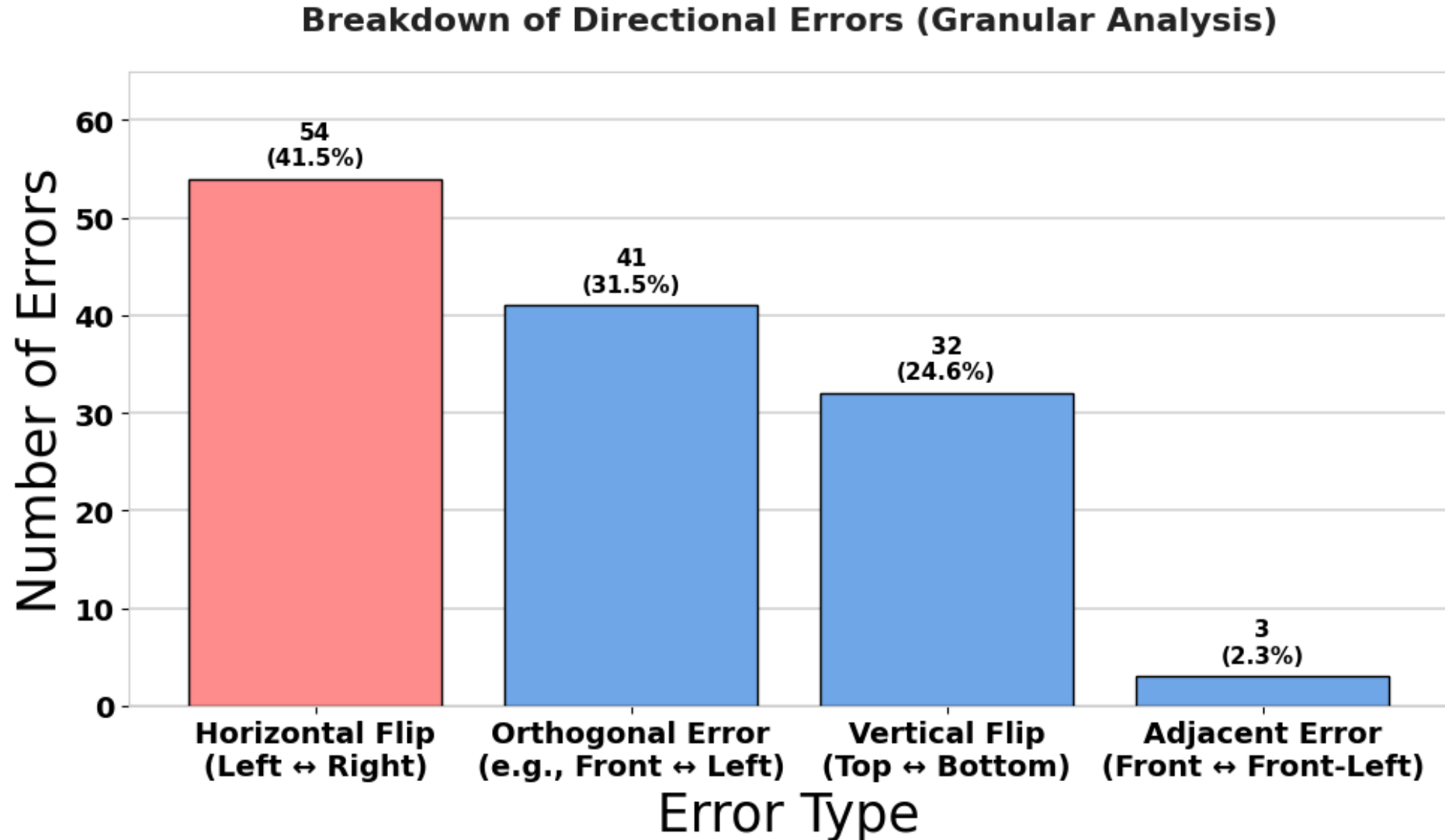
Hypothesis 1 : Data augmentation

- As demonstrated in Vision-Language research, focusing on **learning hard negatives** that confuse the model is key to improving performance, rather than simply augmenting data.
- Hard negative :
 - The most common incorrect answer the model gets wrong
 - image-text pair that is semantically similar but differs in fine-grained details.

Hypothesis 1 : Data augmentation

- **Horizontal Errors:**
 - (Front/Back)Left & (Front/Back)Right
- **Orthogonal Errors:**
 - Front & Left/Right
- **Vertical Flip:**
 - Top & Bottom
- **Adjacent Error:**
 - Angles that are not 90 or 180 degrees
 - Front & Front-left

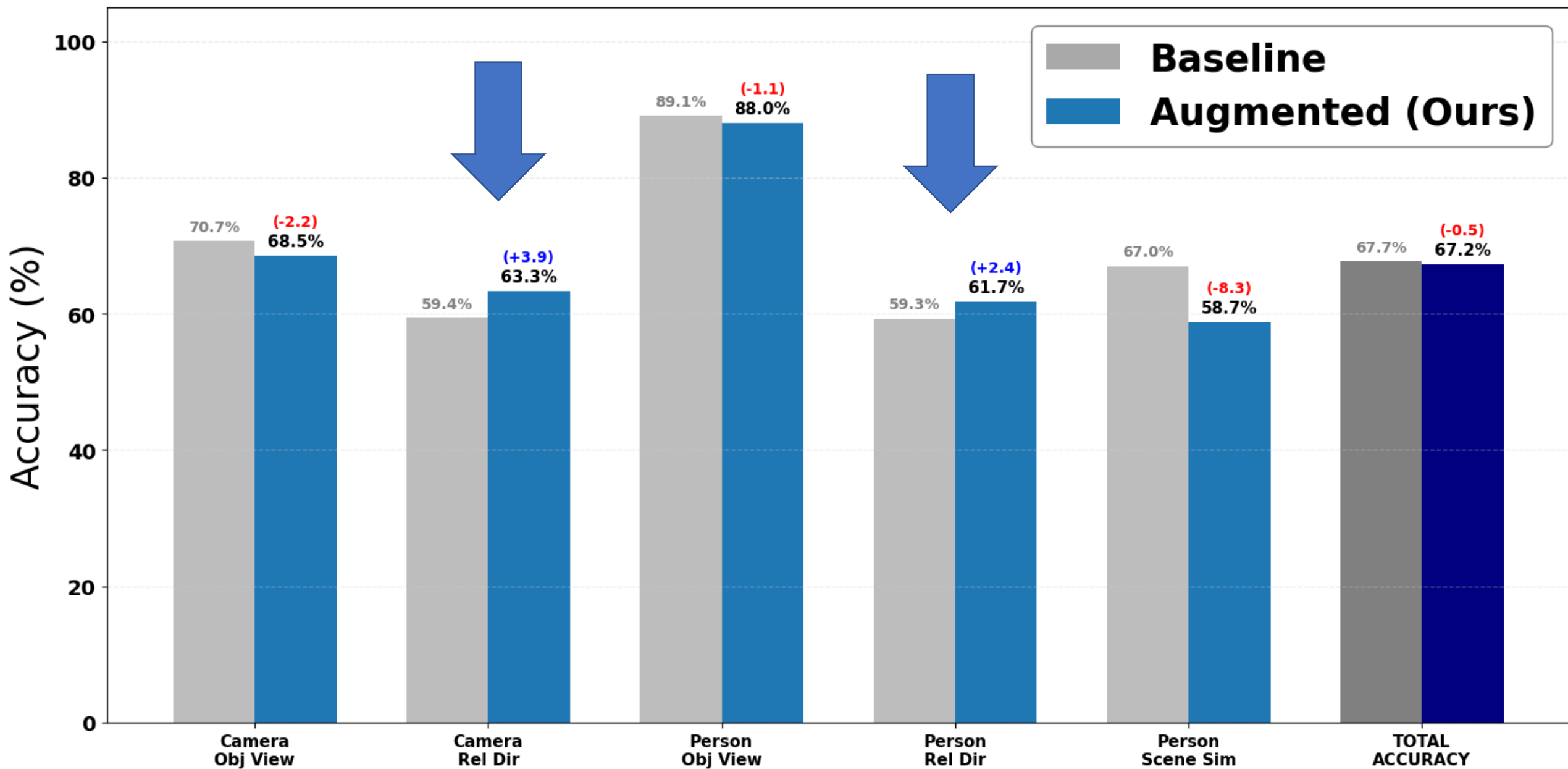
Problem 1 : Direction analysis



Hypothesis 1

- Augment Train dataset
 - Flip the image, flip the label(left , right)

Benchmark Performance Comparison (Baseline vs Augmented)



Result 1 : Data Augmentation

- Although we achieved accuracy improvement in the targeted relative direction,
- We confirmed that the performance of other tasks actually decreased.
 - Which is called **Catastrophic Forgetting***
 - -> To solve this, we can try “cognitive replay”
 - It is a method of repeatedly learning previous data.

Hypothesis 2

Bounding box + Visual prompting

Problem 2 : Object detection fail



☐ **WRONG**

[Question]

From the perspective of the boy, where is the TV located?

[Options]

A. back-left B. front C. right D. left

Problem 2 : Object detection fail

- The question is about “person perspective” but the model is likely looking at "the entire image"

Hypothesis 2 : Bounding box + Visual prompting

- The model may not be good at recognizing **reference objects**.
 - “From the perspective of **the boy**, where is the TV located?”
- -> Then let's make the reference point recognition more certain!
 - Maybe Bounding box might be helpful!
 - Used “Ultralytics YOLO-World*”

How does “Yolo world” works?

- Existing YOLO (v5, v8, etc.) could only find fixed, pre-trained classes (e.g. 80 classes including people, cars, dogs, cats, etc.).
- 1. Text Encoder : Convert user-entered text (e.g., "cat") into vector that the computer can understand. This process utilizes CLIP, a large-scale language-image model, to extract the meaning of the text.
- 2. Image Encoder : Analyzes the input image to extract visual features
- 3. Fusion : If a specific area of the image has a high similarity to the text vector, we determine that it is the object the user is looking for and draw a bounding box there.

Example



MS-CoCo Example

From the perspective of the person wearing the grey clothes, where is the person wearing the blue clothes?



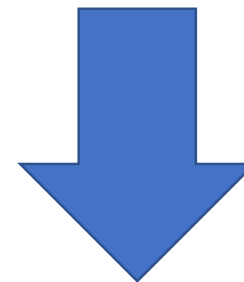
From the perspective of the person **in the red bounding box** wearing the grey clothes, where is the person wearing the blue clothes?

Example



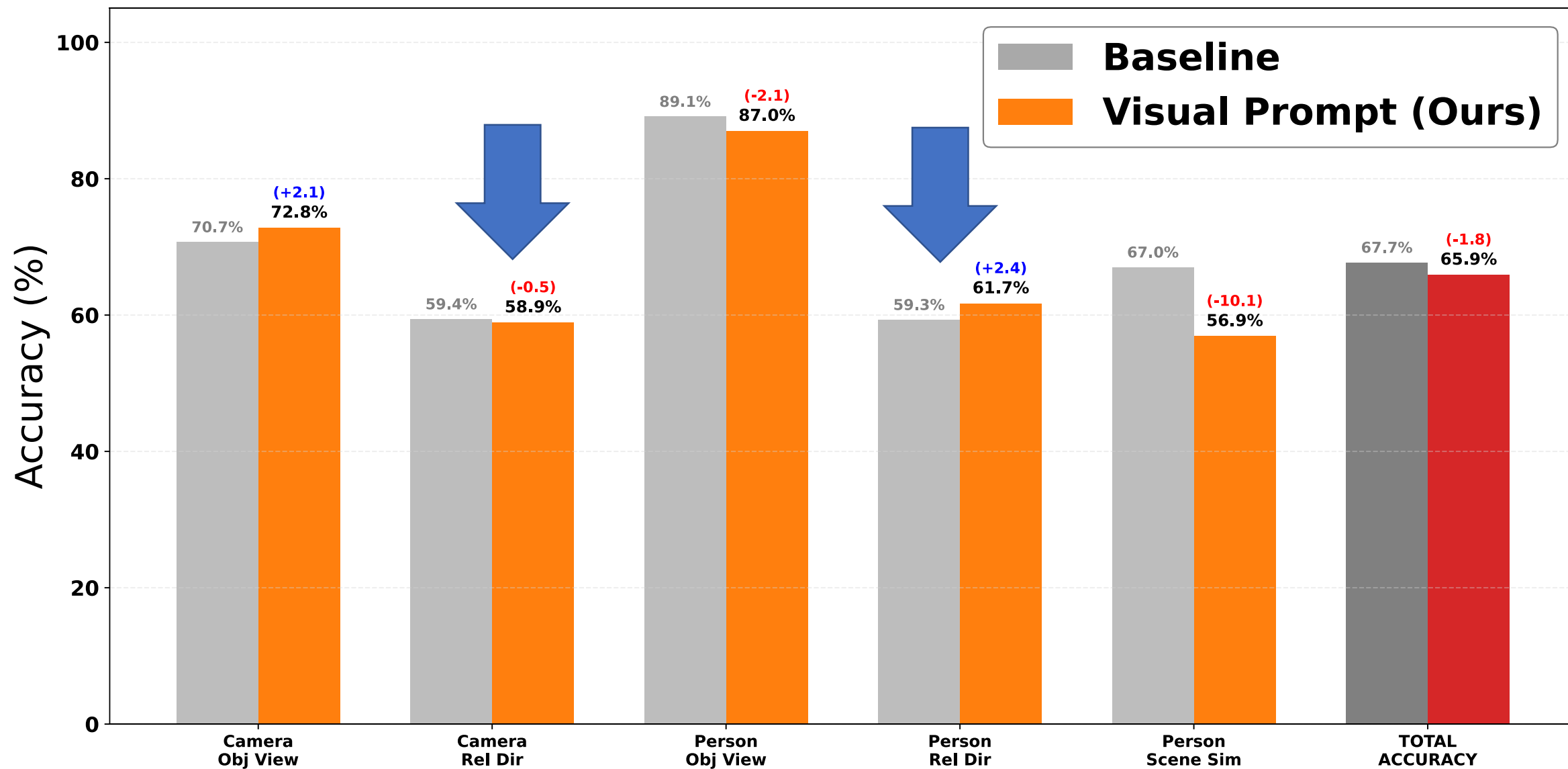
ScanNet Example

How is the sofa positioned with respect to the table?



How is the sofa positioned with respect to the table in the **red bounding box**?

Performance Comparison: Baseline vs Visual Prompting



Result 2 : Bounding Box

- Result :
 - [Per] Rel dir (+2.4%), [Camera] Obj ori(+2.1%): Performance improved in problems where the anchor point must be clear.
- Limitations:
 - [Per] Scene simulation (−10.1%) : The surrounding environment (walls, obstacles, spatial structure) must be grasped as a whole, but this is interpreted as a kind of **tunnel vision** phenomenon where the model's gaze is trapped within a red box.

Do bounding boxes induce tunnel vision?

$$S_i = \left| \frac{\partial f(x)}{\partial x_i} \right|$$

$f(x)$: Logit of prediction
 x_i : pixel value

- Used Saliency map
 - A saliency map is a way to visualize how sensitive each element of the input is to the prediction by calculating the gradient of the input x with respect to the model output y .
 - “If this pixel value changes even a little, how much does the correct answer (Logits) change?”
- Why not attention map?
 - It shows where the model "routed" during its computation, doesn't guarantee that it influenced the correct answer.

[Rel Dir] From the perspective of the man who is looking at the computer, where is the man wearing a hat positioned?



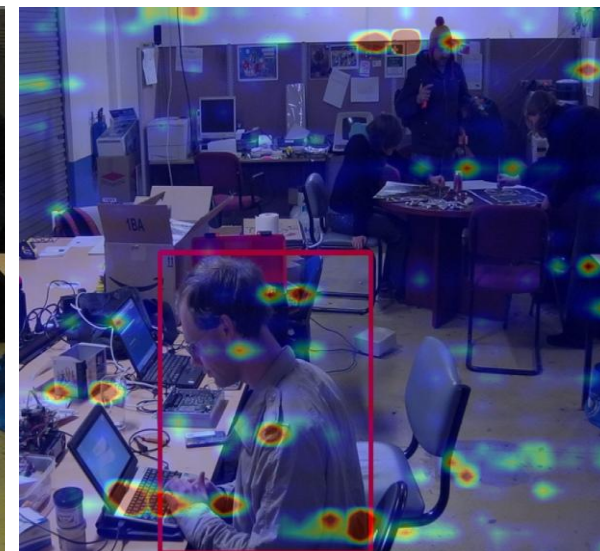
Original Image



Original Saliency



Visual Prompt Image



Boxed Saliency

[Rel Dir] From the perspective of the woman wearing the green helmet, where is the person in black clothes?



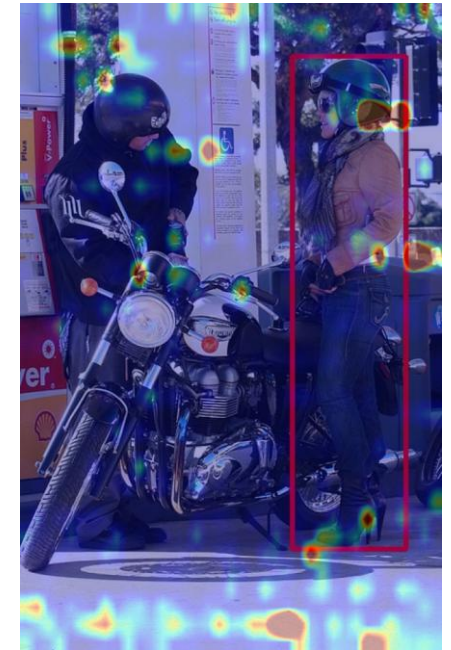
Original Image



Original Saliency



Visual Prompt Image



Boxed Saliency

Hypothesis 3

Chain of thought

Hypothesis 3 : Chain of Thought



Q : With the camera's viewpoint as the front, which direction is the elephant facing in the image? A. right B. front C. back-left D. left
Answer with the option letter.

“A”



Let's think step by step to determine the spatial relationship.

- 1. First, I identify the reference point:** the camera viewpoint.
- 2. Next, I locate the target object:** man.
- 3. By analyzing their relative positions in the 3D space,** man is positioned to the right of the camera viewpoint.

Therefore, the correct option is A.

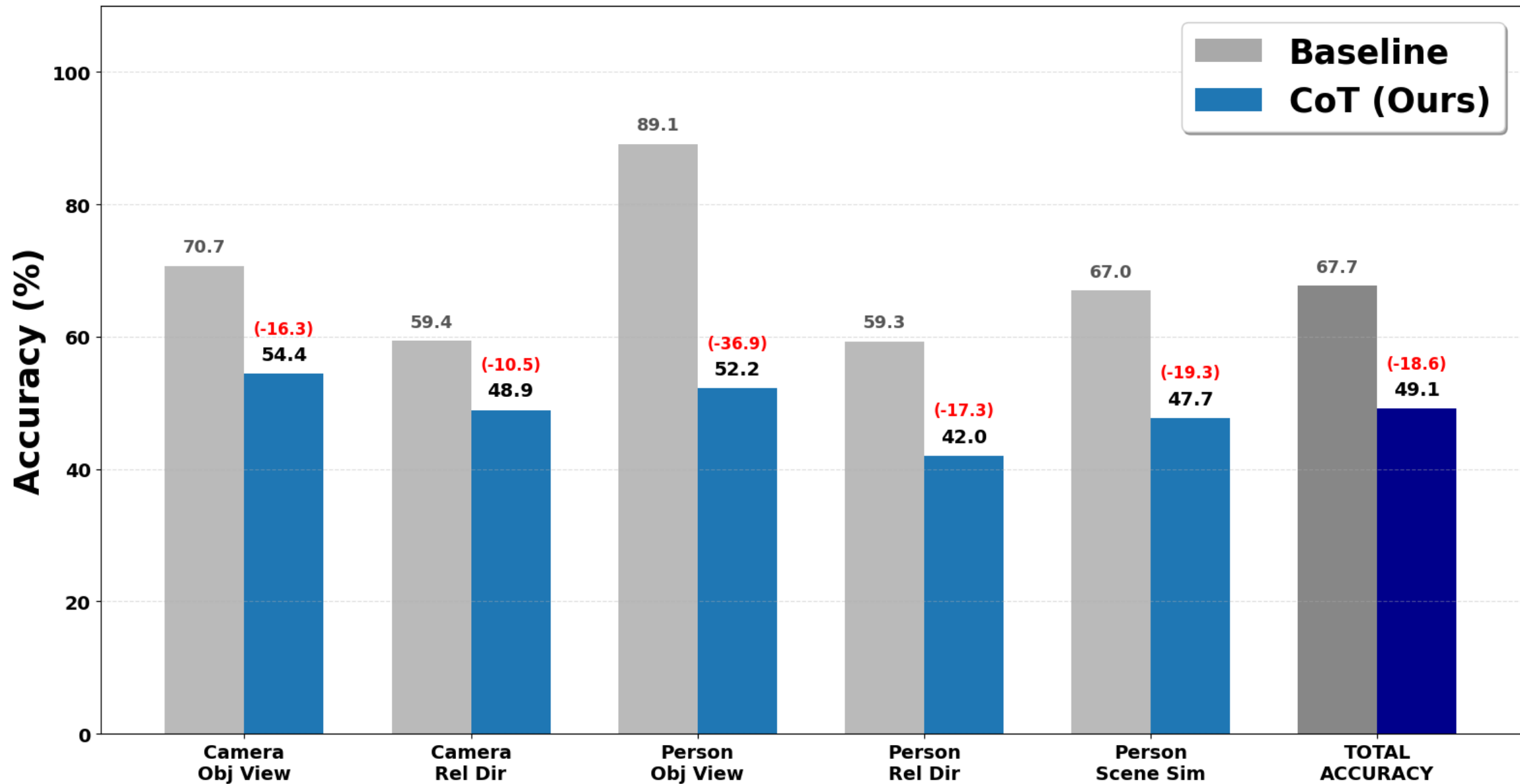
How can we teach chain of thought?

- Use Distilling the knowledge

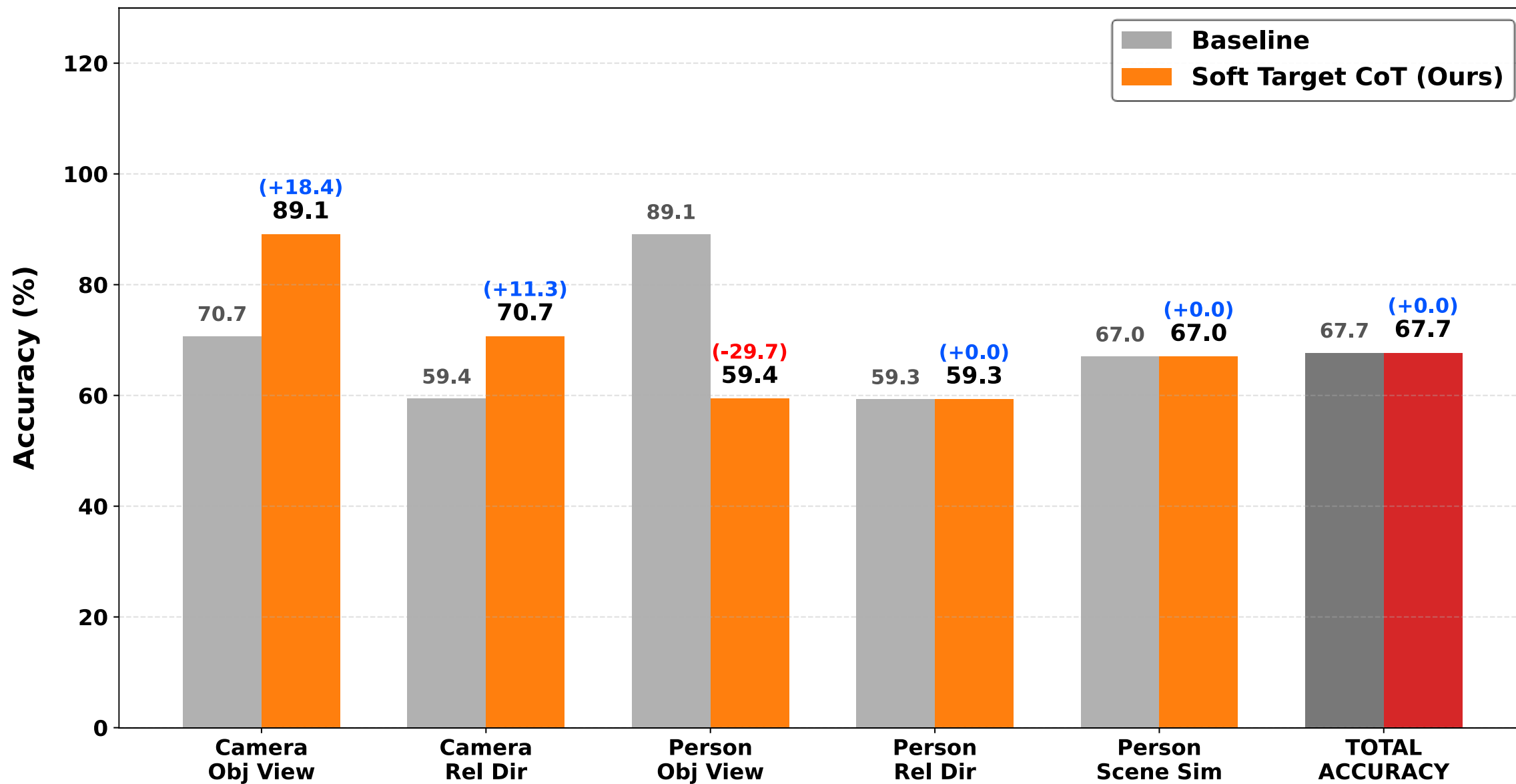
Distilling the knowledge

- Used Qwen 2.5 **7B** model for **teacher** , and our Qwen 2.5 3B model will be a student
 - Qwen2.5 73B model was unusable due to OOM issue.
- Hard target : train with one-hot vector
 - Ex) [0,0,1]
- Soft target : train with prob. Distribution(after softmax)
 - Ex) [0.1, 0.2, 0.7]

Benchmark Performance: Baseline vs CoT(w Hard target)



Performance Comparison: Baseline vs Soft Target CoT



Result 3 : CoT

- Why did the results turn out this way?
 - Problem : Consistent decline from a person perspective
 - Inductive Bias transferred from the Teacher Model (Qwen 2.5 7B)
- Limitations : Used the Qwen 2.5 **7B** model instead of the 72B model or other higher end models.
 - How can we improve performance even further?
 - Better open source model (InternVL2-26B, Llama-3.2-11B-Vision, Qwen3)

Conclusion

- We confirmed that **data augmentation(for hard negatives)** and **bounding boxes**, and **CoT** helped improve performance

Thank you!

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